

CLINICAL AND POPULATION STUDIES



Proteogenomic Analysis of Coronary Artery Calcification in Human Populations

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BACKGROUND: Joint use of multiple molecular layers can be useful to prioritize targets for mechanistic studies. Application of this approach to coronary disease in large populations is an emerging field.

METHODS: We used reported circulating proteomic data (Somascan aptamer-based) from ≈3000 individuals in the CARDIA study (Coronary Artery Risk Development in Young Adults), measuring association with prevalent and 10-year incident coronary artery calcium (CAC) score. We used a multiparametric approach to prioritize circulating protein-CAC associations via genomics of circulating protein levels and coronary artery transcription.

RESULTS: Proteins linked to prevalent/incident CAC in CARDIA implicated pathogenic mechanisms of vascular disease, including fibrosis and inflammation (GDF-15 [growth/differentiation factor 15], CDCP1 [CUB domain-containing protein 1], GSN [gelsolin], THBS2 [thrombospondin-2], chemokines, RNASEH1 [ribonuclease H1]), oxidative lipid metabolism (CILP2; cartilage intermediate layer protein 2), extracellular matrix remodeling and signaling (MMPs [matrix metalloproteinases], TIMP-1 [tissue inhibitor of metalloproteinases 1], integrins), calcification (Notch 1, ARHGAP36 [Rho GTPase-activating protein 36]), and metabolism (GIP [gastric inhibitory polypeptide]), as well as new proteins not previously reported. Using proteome-wide association study genetic approaches, several targets with nominal evidence in CAC proteomics were associated with atherosclerosis or myocardial infarction in over 300K individuals, including PCSK9 (proprotein convertase subtilisin/kexin type 9) and APOC1. Finally, the coronary artery-specific transcriptome-wide association study of CAC yielded genes with previously implicated mechanistic roles in vascular homeostasis, inflammation, and metabolism, as well as genes without previously described function in CAC. Overlap across CAC proteomics and transcriptome-wide association study highlighted genes involved in vascular inflammation (S100A9), cardiac development (HES1 [transcription factor HES-1]), vessel wall structure (SPARCL1 [SPARC-like protein 1]), and vascular dysfunction or plaque (NOTCH3 [neurogenic locus notch homolog protein 3], TNFSF12 [tumor necrosis factor ligand superfamily member 12], S100A12 [protein S100-A12]).

CONCLUSIONS: These results report population-level multiomics in human coronary calcification, presenting a method to identify disease-relevant targets through integration of human genetic approaches with multiomics.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: coronary disease ■ inflammation ■ multiomics ■ proteomics ■ risk

Atherosclerotic cardiovascular disease (CVD) is the most common cause of cardiovascular demise worldwide.¹ CVD has origins decades before the

occurrence of key clinical events, including sudden cardiac death and myocardial infarction.² Despite ongoing, broad pharmacological and behavioral efforts to mitigate

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Nonstandard Abbreviations and Acronyms

ARHGAP36	Rho GTPase-activating protein 36
CAC	coronary artery calcium
CARDIA	Coronary Artery Risk Development in Young Adults Study
CDCP1	CUB domain-containing protein 1
CVD	cardiovascular disease
FHS	Framingham Heart Study
GDF-15	growth/differentiation factor 15
GIP	gastric inhibitory polypeptide
GSN	gelsolin
GTEX	genotype-expression tissue database
GWAS	genome-wide association study
HDL	high-density lipoprotein
JTI	joint-tissue inference
MMP	matrix metalloproteinase
MR	Mendelian randomization
PCSK9	proprotein convertase subtilisin/kexin type 9
PQTL	protein quantitative trait loci
PWAS	proteome-wide association studies
THBS2	thrombospondin-2
TWAS	transcriptome-wide association study
UBE2G2	ubiquitin-conjugating enzyme E2 G2

risk, progressive worsening in dysmetabolic phenotypes has promoted an increase in early CVD, with nearly 50% of CVD mortality under age 65 years in some reports.^{3,4} Accordingly, defining underlying targetable CVD mechanisms at an early subclinical stage is a major goal in modern precision cardiovascular prevention. Coronary artery calcification (CAC)—a touchstone phenotype of early CVD susceptibility with therapeutic implication⁵—represents a culmination of interconnected molecular events within the endothelium and the vascular wall involving local and systemic inflammation, immune activation, and metabolic-oxidative stress.⁶ Molecular studies of early coronary disease progression in humans have focused on relating accessible genetic or metabolic states (via metabolites, proteins, peripheral blood transcriptomes)^{7–9} to CAC or clinical CVD. Nevertheless, examining underlying determinants of coronary artery disease—including in a potentially tissue-specific fashion, limited due to inaccessibility of coronary tissue⁶—is critical to identify and interrupt early, intervenable changes in human CVD.

Here, we integrate proteomics and genomic techniques toward 2 major goals: (1) to identify circulating proteins related to the presence, extent, and progression of CAC; (2) to map these targets into human coronary artery tissue through human genetic and transcriptional techniques. For the former (and primary) goal, we identified circulating proteins associated with both prevalent

What Are the Clinical Implications?

This study presents a multilevel proteogenomic framework that integrates population-scale circulating proteomics with coronary artery-specific transcriptomics to prioritize targets for precision cardiovascular prevention. By indexing the proteome to both prevalent and 10-year incident coronary artery calcium, we identify molecular mediators of subclinical disease that may appear decades before clinical events. We report the largest human coronary artery transcriptome-wide association study of coronary artery calcium to date, identifying high-priority protein-gene axes—such as NOTCH3 (neurogenic locus notch homolog protein 3), S100A9 (protein S100-A9), and HS6ST3 (heparan-sulfate 6-O-sulfotransferase 3)—that operate within the vessel wall and the systemic circulation. Our findings demonstrate that integrating tissue-specific genomics with circulating proteomics may bridge the gap between risk association and causal mechanism. For instance, the identification of targets like PCSK9 (proprotein convertase subtilisin/kexin type 9) and APOC1 through proteome-wide association validates the framework’s ability to recover established therapeutic benchmarks, while the discovery of novel targets in extracellular matrix remodeling and inflammatory signaling offers a roadmap for future drug development. Ultimately, this work provides a biologically curated resource to transition from descriptive risk prediction to mechanism-based discovery, facilitating the identification of biomarkers and therapeutic targets aimed at interrupting the progression of calcific atherosclerosis in its earliest stages.

and 10-year incident CAC phenotypes in ≈3000 individuals in the CARDIA study (Coronary Artery Risk Development in Young Adults), with validation in the FHS (Framingham Heart Study; both using extant data). For our secondary goal, we examined whether targets identified at the protein level have evidence for potential causal roles in CAC and downstream CVD. To accomplish this, we used 2 parallel approaches: (1) genetic approaches targeting the circulating proteomic correlates of CAC (via Mendelian randomization [MR] and proteome-wide association studies [PWAS]) and (2) genetic approaches targeting gene expression in human coronaries (transcriptome-wide association study [TWAS] of CAC in >35 000 individuals based on genetic models to predict gene expression in human coronary arteries from the GTEx [Genotype-Expression Tissue Database] consortium). These approaches use human genetic variants (or models based on multiple variants) associated with circulating proteins (MR and PWAS) or coronary artery transcripts (TWAS) to perform an association study with the phenotype or clinical outcome of interest; given that the genetic variants are assigned in meiosis, these results

are less susceptible to confounding than proteomic associations. As with our proteomic findings, we present our PWAS and TWAS results as a scientific resource for further discovery. Although transcriptional events in the coronary wall are not necessarily reflected in the circulating proteome, we explored potential overlap across the TWAS and proteomic associations of CAC in human coronary arteries with and without atherosclerosis at a single-cell level to identify potential concordances that may further parse targets for future mechanistic validation. Ultimately, this work reports a proteogenomic analysis of liability to coronary calcification—an important harbinger of clinical vascular events—identifying both circulating and tissue-relevant determinants of CAC for therapeutic and biomarker validation in future studies.

MATERIALS AND METHODS

All data and materials have been made publicly available at the CARDIA Coordinating Center (Coronary Artery Risk Development in Young Adults – Heersink School of Medicine; <https://sites.uab.edu/cardia>) and can be accessed via email to coc@uab.edu. No new code was developed as part of this study. Data from the FHS is available at Database of Genotypes and Phenotypes (dbGaP; dbGaP identifier ph006013) or via contact with the FHS coordinating center (www.framingham-heartstudy.org). UK Biobank data (accessed under application number 94960) is available at the UK Biobank Research Access Portal. Code used for analysis in this study are available on Zenodo (for TWAS models: <https://doi.org/10.5281/zenodo.3842289>) and on the GitHub repository for this project (<https://github.com/gamazonlab/CAComics>).

Clinical Cohorts

Coronary Artery Risk Development in Young Adults

A total of 2971 participants with circulating proteomics and CAC scores by computed tomography at the year 25 visit were included.¹⁰ Clinical-demographic data were collected by standardized assessment as described,^{11,12} and computed tomography for CAC was performed as previously noted via standardized protocols.^{13,14} In addition, 793 participants had additional computed tomographic CAC data collected at year 35 in CARDIA for analysis of incident CAC development by year 35 (in those individuals without CAC at year 25). Study participants provided written informed consent, and the study was approved by the respective clinical site institutional review board.

Framingham Heart Study

We studied 573 individuals with extant proteomics and computed tomographic CAC measurements from the FHS Generation 2 (Offspring) cohort to assess for replication of our findings in CARDIA.¹⁵ CAC was performed a median of 17.4 years after collection of the blood sample in FHS (which was eventually used for proteomics). Methods for clinical and calcification traits have been reported.^{16,17} All FHS participants gave written informed consent, and all study protocols received approval from the institutional review board at Boston University Medical Center.

Quantification of the Circulating Proteome

In CARDIA, we used the SomaScan proteomics platform (aptamer-based technology) to quantify 7228 aptamers representing 6402 unique proteins in CARDIA (Somalogic, Boulder, CO), as reported in previous work.¹⁰ We excluded 71 aptamers (41 unique proteins) with a coefficient of variation exceeding 20%. After log₂ transformation and standardization of aptamer levels, aptamer levels were winsorized to limit the influence of extreme values. Specifically, standardized aptamer values exceeding 5 or −5 (ie, ±5 SDs from the mean, calculated per aptamer) were recoded to 5 or −5, respectively.

To explore the generalizability of our findings, we studied population-level proteomics in FHS. In FHS, protein profiling was completed in 2 batches. In batch 1, 1129 aptamers were profiled in 821 individuals, and batch 2 included an expanded panel of 1372 aptamers, which was assayed in 1092 participants. Aptamer levels of 1128 aptamers common to both batches were log-transformed, standardized within batch, then pooled and normalized to a mean value of 0 and SD unit of 1 using the Blom rank-based inverse-normal method. These values were then regressed on assay plate ID to account for plate effects, and the standardized residuals were used for regression. Methods for handling proteomic data (batch pooling, transformations, and rank normalization) have been reported¹⁸; we have reproduced them with minimal change to maximize reproducibility. Matching of CARDIA to FHS aptamers was performed by SomaScan seqID (1035 overlapping aptamers representing 1010 unique proteins by UniProt identifier).

Associations of the Circulating Proteome to CAC

In CARDIA, protein aptamers were log-transformed, centered, standardized to unit variance, and winsorized to 5 SDs for regression. Given the absence of known data sets with a harmonized proteomic platform and concurrent measures of CAC, we randomly split our CARDIA sample into a derivation (70%) and validation subsample (30%). We estimated models in the discovery subsample for the presence/absence of CAC (as a binary variable in logistic models) and its extent (continuous in linear models; modeled as natural log(CAC+1) (ln[CAC+1]) to account for zero CAC values). Although the ln(CAC+1) transformation does not fully normalize the distribution, it is a commonly used approach in prior studies to accommodate the skewed nature of CAC data for statistical modeling.^{19–21} Each model contained one aptamer at a time, adjusted for age, sex, and race (Black or White), with a 5% false discovery rate (FDR; Benjamini-Hochberg FDR) applied across models to control for multiplicity. Aptamers that were significant at a 5% FDR threshold were passed to identical models in our validation subsample. Aptamers that were significant at 5% FDR for both the presence of CAC and its extent in both derivation and validation subsamples were analyzed in similar logistic regression models (adjusted for age, sex, and race) for incident CAC among subjects who had CAC scores of 0 at year 25 and available computed tomography data at year 35. Aptamers in the incident CAC models were considered significant using a threshold of 5% FDR. We included all aptamers, regardless of duplication, as a transparent scientific resource for replication for the community. Proteomic associations with prevalent CAC (year 25) were performed using SAS. Proteomic associations

with incident CAC (year 35) and data visualization of all proteomic associations were performed using R 4.3.1 (packages ggplot2_3.4.3, ComplexHeatmap_2.18, circlize_0.4.16).

Replication studies in FHS were performed in a similar fashion (logistic for presence or absence of CAC; continuous linear for extent as standardized $\ln[\text{CAC}+1]$), with each protein in separate models adjusted for age, sex, and race (White versus non-White), with 5% FDR used to control type 1 error. We assessed the concordance of the CARDIA and FHS effect size via Pearson correlation.

Genetic Inferential Approaches Based on Circulating Proteomic Associations With CAC

MR Approaches for the Circulating Proteome

We identified protein quantitative trait loci (pQTLs, defined here as genetic instruments associated with circulating protein levels at genome-wide significance) that were measured with 4907 aptamers in 35559 Icelanders (summary statistics obtained from a prior study²² at decode.com/summarydata/; quality control reported previously). We conservatively selected pQTLs exceeding a genomic threshold ($P < 5 \times 10^{-8}$) for the target proteins, leveraging both cis-pQTLs (+2 Mbp on both directions of the encoding locus) and trans-pQTL (when available). Although trans-pQTLs are more likely to regulate multiple targets pleiotropically, for the purpose of MR, inclusion of trans-pQTLs may improve protein-phenotype associations as they can increase MR power analysis.²³ This subset was subject to linkage disequilibrium clumping ($R^2 < 0.001$, 10 kbp window, 1000 Genome European reference). We restricted pQTLs to those exceeding an F statistic over 10 to minimize downstream weak-instrument bias. Of 131 unique proteins (by UniProt identifier) that were associated with the presence and extent of CAC at a 5% FDR in the derivation and validation samples in CARDIA, we identified at least 1 pQTL for 100 proteins. For disease states, we selected 3 of the largest genome-wide association studies (GWASs) capturing CAC-relevant clinical disease states for assessment in MR: (1) coronary calcification (N=36720 Europeans; GWAS catalog accession GCST90278455)²¹; (2) coronary artery disease (N=1165690 Europeans; GWAS catalog accession GCST90132314)²⁴; (3) myocardial infarction (N=639221 Europeans; GWAS catalog accession GCST011365).²⁵

Our 2-sample MR framework tested whether proteins implicated in our initial epidemiological and tissue evaluation had evidence of potential causality by human genetic approaches to CAC and relevant outcomes. We used pQTLs (as instrumental variables for genetically determined protein levels) as exposures and our 3 CAC-relevant clinical disease states as outcomes. Quality control was performed using TwoSampleMR R 0.6.21 package² with correction for strand mismatches via the `harmonise_data()` function. All primary variants present in both the outcome and predictor data sets were considered for the analysis; proxy variants were not included. Compatibility of genome builds between the summary statistics used for the outcome and predictors (pQTLs) was confirmed before analysis. We primarily tested potential causal associations with the inverse variance-weighted robust-penalized method, using the MendelianRandomization 0.10.0 R package.³ We used the default effect model for association testing, or the random effect model when instrumental variables for the protein demonstrated significant heterogeneity in effect size (Q -het

$P < 0.05$). Given that the candidate targets that we passed on to MR analyses already had evidence from our proteomic association studies, we considered the protein as potentially causally associated with the CAC-relevant outcome using a nominal $P < 0.05$ threshold. In addition, we used median penalized (to limit instrumental variable-related bias) and Egger-penalized robust methods (for pleiotropy) as sensitivity analyses to enhance our confidence in causal interpretations, consistent with STROBE-MR (Strengthening the Reporting of Observational Studies in Epidemiology Using Mendelian Randomization) guideline for causal inference using observational data.²⁶ Our MR analyses were performed in R 4.4.0.

Proteome-Wide Association Studies

As a secondary approach, we performed PWAS of CAC, using genetic models of the circulating proteome developed in a large-scale proteogenomic study (N=7213, European ancestry; 4657 aptamers corresponding to 4435 unique genes).²⁷ We included only those proteins that attain nominal significance during model training ($P < 0.05$, cross-validation $r = 0.10$, 1340 unique proteins). This resource does not overlap with the UK Biobank (N=361194), allowing us to estimate the association between the genetically determined circulating protein expression and related traits in an independent phenomic resource. We used GWAS summary statistics from a linear model of a phenotype as a function of the first 20 genotype-based principal components, sex, age, age², sex×age, sex×age², and the protein under test to estimate the effect size of the protein on the 2 phenotypes in the UK Biobank: coronary atherosclerosis (code I9_CORATHER) and myocardial infarction (code I9_MI). We tested proteins associated with the presence and extent of CAC in CARDIA (derivation $P < 0.05$) with unique gene symbols in PWAS.

Genetic Inferential Approaches Based on the Coronary Artery Transcriptome

We developed gene expression models in human coronary artery samples from 268 unrelated individuals in the GTEx database. This approach uses extant (but not previously widely used) bulk RNA-sequencing transcriptomic data in human coronary arteries alongside whole genome sequencing data in GTEx to develop genetic models of transcript expression that can be applied more generally to allow genetic inference in large GWAS or other genomic resources. We used our joint-tissue inference (JTI) TWAS methodology, which has been reported and validated.²⁸ JTI is an approach to test the effect of genetically determined gene expression on phenotype. JTI consists of a set of gene expression models using cis genetic variants as features. The training of models is improved by the use of shared regulatory architecture between tissues through joint-tissue analysis of a unique multitissue (49 tissues) transcriptome resource (GTEx). The approach also integrates a causal inference methodology on the effect of gene expression on the phenotype under study. The JTI models can be applied to GWAS data to identify gene-phenotype associations. The training data used to develop the JTI models was GTEx (v8). We trained prediction models of normalized gene expression using sex, platform, and 5 genotype-based principal components as adjustments. We also performed phenome-wide association studies by identifying the associations of the genes with the heritable phenome (heritability h^2 $P < 0.05$) as captured in the UK Biobank.

Single-Cell RNA Sequencing of Human Coronary Arteries

We leveraged single-cell RNA sequencing data derived from 13 human coronary arteries (8 with atherosclerotic lesions and the remainder lesion-free controls) involving 56 183 cells.²⁹ We analyzed the data using the Seurat R 5.2.1 package.³⁰ Doublet removal was performed using scDblFinder 1.18.0.³¹ scDblFinder used 3 iterations, and only consensus cell IDs from all 3 runs were retained as (consensus) doublets (following Mosquera et al²⁹). Cell barcodes identified as doublets were then filtered out from the raw counts matrix. As ambient RNA contamination can negatively affect gene expression profiling, we performed correction via DecontX using default parameters.³² We applied raw count normalization through a regularized negative binomial regression approach as implemented in SCTransform 0.4.1³³ (with argument `vst.flavor=v2`). To avoid confounding by cell cycle state, we adjusted for cell cycle variance. For integration of processed sequencing libraries across potential technical differences and data set–specific conditions, we used Harmony 1.2.3.³⁴ We applied dimensionality reduction using principal component analysis on the normalized counts, leveraging the first 30 principal components for clustering. We invoked the functions RunUMAP (using 30 neighboring points for local approximation of manifold structure) and FindNeighbors (using $k=20$ for the k -nearest neighbor algorithm) in Seurat, setting reduction to harmony to use the Harmony embeddings. Cell type annotation leveraged Seurat’s reference-based transfer learning applied to the Tabula Sapiens (specifically, vasculature data in 42 650 cells available at <https://cellxgene.cziscience.com/e/a2d4d33e-4c62-4361-b80a-9be53d2e50e8.cxg/>).³⁵ For Tabula Sapiens, we applied the same SCTransform normalization workflow and UMAP dimensionality reduction as used for the samples, ensuring consistency with the cell annotations. We examined differential expression between the samples with atherosclerotic lesions and the lesion-free samples using pseudobulking-based DESeq2.³⁶ We used 5% FDR to control type 1 error.

RESULTS

Characteristics of Proteomic Discovery Cohorts

Our overall study scheme is shown in the Graphic Abstract. Clinical characteristics of the study populations (CARDIA: N=2971; FHS: N=573) are in Tables 1 and 2. CARDIA participants at year 25 had a mean age of 50.2±3.6 years, with approximately even distribution by sex and self-identified race (56% women, 46% Black) and prevalent cardiometabolic risk (mean systolic blood pressure, 119±15 mmHg; 8% diabetes; and mean body mass index, 30.3±7.1 kg/m²) and CAC (≈29%). A subgroup of participants with baseline proteomics and without prevalent CAC at year 25 had computed tomography performed ≈10 years later (Year 35 study visit in CARDIA; N=793), with an incident CAC rate of ≈31%. FHS participants were similar in age and sex, with a low prevalence of clinical CVD (≈4%) at the time of proteomics, with a high rate of CAC (≈69%; when measured ≈20 years after the blood sample was collected for proteomics).

Table 1. Characteristics of Study Samples (Data From CARDIA Study From the Year 25 Examination)

Covariate	CARDIA derivation (n=2087)	CARDIA validation (n=884)
	Mean (SD)/n (%)	Mean (SD)/n (%)
Age, y	50.1 (3.6)	50.3 (3.6)
Female sex	1166 (55.9%)	495 (56.0%)
Black	961 (46.0%)	419 (47.4%)
Body mass index, kg/m ²	30.3 (7.1)	30.1 (7.2)
Systolic blood pressure, mm Hg	118.8 (15.4)	118.2 (15.2)
Smoking status		
Current smoker	570 (27.5%)	221 (25.1%)
Quit within a year	111 (5.4%)	60 (6.8%)
Never or quit over a year	1392 (67.1%)	598 (68.0%)
Diabetes (fasting glucose ≥126 mg/dL or on medications)	166 (8.0%)	74 (8.4%)
Total cholesterol, mg/dL	192.4 (36.9)	191.3 (36.6)
Total high-density lipoprotein cholesterol, mg/dL	57.5 (17.8)	58.3 (17.9)
Total low-density lipoprotein cholesterol, mg/dL	112.1 (32.5)	111.0 (33.3)
Taking cholesterol-lowering medication	328 (15.8%)	150 (17.0%)
Any CVD disease by year 25	65 (3.1%)	32 (3.6%)
Coronary artery calcium, AU	48.0 (210.9)	48.6 (265.6)
Coronary artery calcium >0, AU	634 (30.4%)	222 (25.1%)

AU indicates Agatston units; CARDIA, Coronary Artery Risk Development in Young Adults; and CVD, cardiovascular disease.

Identifying and Characterizing the Circulating Proteomic Architecture of CAC

We identified 136 aptamers (representing 131 unique proteins) associated with the presence and extent of CAC across discovery and validation sets within CARDIA (Figure 1A; for example, volcano plot for CAC extent; presence of CAC, 186 aptamers, 174 unique proteins; extent of CAC, 201 aptamers, 193 unique proteins; and regression summaries in Table S1). Of note, proteins with multiple aptamers significant against either extent or presence of CAC had concordant directionality across aptamers. Identified proteins exhibited strong biological plausibility across known mechanisms of vascular remodeling (Table 3), including fibrosis and inflammatory mechanisms (GDF-15 [growth/differentiation factor 15],⁵⁶ CDCP1 [CUB domain–containing protein 1],⁴⁰ GSN [gelsolin],³⁹ THBS2 [thrombospondin-2],⁵⁷ chemokines, RNAS6 [ribonuclease K6]⁴⁵), oxidative lipid metabolism (CILP2 [cartilage intermediate layer protein 2]⁵⁸), extracellular matrix remodeling and signaling (MMP [matrix metalloproteinase]-7, MMP-12,⁵⁹ TIMP-1 [tissue inhibitor of metalloproteinases 1], integrins⁶⁰), calcification (Notch 1,⁶¹ ARHGAP36 [Rho GTPase-activating protein 36]⁴⁴), and general metabolism (GIP [gastric inhibitory polypeptide]⁶²). Importantly, a majority of identified

Table 2. Characteristics of Study Samples (Data From FHS)

Covariate	n=573; mean (SD)/n (%)
Age, y	51 (9)
Female sex	339 (59.2%)
Non-White	2 (0.3%)
Body mass index, kg/m ²	27 (5)
Current smoker	67 (11.7%)
Systolic blood pressure, mm Hg	122 (17)
Treated for hypertension	47 (8.2%)
Diabetes	17 (3.0%)
Total cholesterol, mg/dL	202 (35)
Total high-density lipoprotein cholesterol, mg/dL	52 (15)
Treated for lipids	26 (4.5%)
Any CVD by exam 5	9 (1.6%)
Any CVD by CT scan date*	55 (9.6%)
CAC >0	395 (69%)
CAC score, AU	324 (705)

AU indicates Agatston units; CAC, coronary artery calcium; CT, computed tomography; CVD, cardiovascular disease; and FHS, Framingham Heart Study.

*CT scan was performed a median of 17.4 y after blood for proteomics was collected.

protein associations (though biologically plausible) had not been previously widely reported in human coronary disease or calcification, including mechanisms of protein catabolism (UBE2G2 [ubiquitin-conjugating enzyme E2 G2]⁴²), macrophage efferocytosis and lipid metabolism (TREM2 [triggering receptor expressed on myeloid cells 2]⁶³), endothelial cell states (EGFR [epidermal growth factor receptor]⁶⁴), and extracellular matrix metabolism (PCOLCE [procollagen C-endopeptidase enhancer]).

To address epidemiological limitations on cross-sectional associations (reverse causation), we quantified the relation of 136 aptamers in our prevalent CAC proteome with incident development of any CAC ≈10 years later in a subsample of CARDIA that was CAC-free at the time of proteomics (N=793; Figure 1B; Table S2). Among aptamers related to the prevalence and extent of CAC at baseline, we observed strong consistency in directionality of effect (129/136) with incident CAC at 10 years (with 59 aptamers significant for incident CAC after FDR correction). In addition, we examined associations with CAC measured ≈20 years after the blood sample (on which proteomics was performed) in FHS, observing broad consistency in effect directionality and size for proteins with significant relationships for CAC in CARDIA (Figure S1; full regression results in Table S3). This includes proteins with complete agreement in effect size and significant relationships in FHS, including MMP-12 (higher levels previously associated with more coronary atherosclerosis^{65,66}) and Contactin-1 (higher levels previously associated with absence of coronary atherosclerosis^{65,66}).

MR Studies of Proteins Associated With CAC

Next, we performed MR studies to increase confidence in the targetability of select mediators and their causal relevance to human CVD. For MR, we used established pQTL for proteins associated with the presence and extent of CAC in CARDIA (100 included that satisfied quality control thresholds; see Materials and Methods). We observed broad consistency between MR-based genetic estimates of effect on 3 major disease conditions (CAC, coronary artery disease, and myocardial infarction; spanning ≈1.8 million individuals) and proteomic-phenotype associations from CARDIA (Figure 1C; full results in Table S4), several of which have not been widely reported in coronary vascular calcification in humans. While a full exposition of findings is out of scope for this work, several findings merit mention. Proteins with both proteomic and genetic evidence in favor of increased calcification or clinical risk (at $P<0.05$ for at least 1 MR method) included mediators involved in central mechanisms of metabolism and inflammation, including *GABARAPL1*⁶⁷ (autophagy), *S100A9*/calgranulin B (inflammatory signaling/atherosclerosis in murine models⁶⁸), *VEGFA*^{69,70} (angiogenesis and lipid handling), *C1QTNF1*⁷¹ (macrophage activation). Several genes with evidence of protection against end points at a proteomic and genomic level had not been broadly implicated in calcification in humans, including proteins implicated in central nervous system development and physiology (*SLITRK1*^{72,73}; noradrenergic system development; *IGLON5*⁷⁴; *NTRK3*), extracellular matrix metabolism (*HS6ST3*, *ADAM23*), and fibroblast development (*FGFR1*).

Circulating Proteome-Wide Association Studies of Proteins Associated With CAC

In a parallel approach, we estimated the association of genetically determined protein abundance with atherosclerotic disease phenotypes in 361 194 UK Biobank participants through proteome-wide genetic association studies (PWAS). Our studies prioritized those proteins associated with CAC in CARDIA proteomics (in derivation $P<0.05$) with suitable PWAS models (see Materials and Methods). At a 5% Bonferroni threshold (across the tested proteins), we observed 4 proteins with significant evidence for involvement of genetically determined protein abundance in atherosclerosis (proprotein convertase subtilisin/kexin type 9 [*PCSK9*], *APOC1*, *HTRA1*, and *FGFR1*; Figure 1D) and 2 for myocardial infarction (*PCSK9*, *INHBC*; Figure 1E; Table S5). Notably, the directly measured CAC-associated proteins in CARDIA were substantially enriched for genetically determined CAC associations by PWAS (Figure 1F), with many of the targets among the most significant associations from PWAS. For example, circulating PCSK9 expression is linked to greater severity and vulnerability of coronary

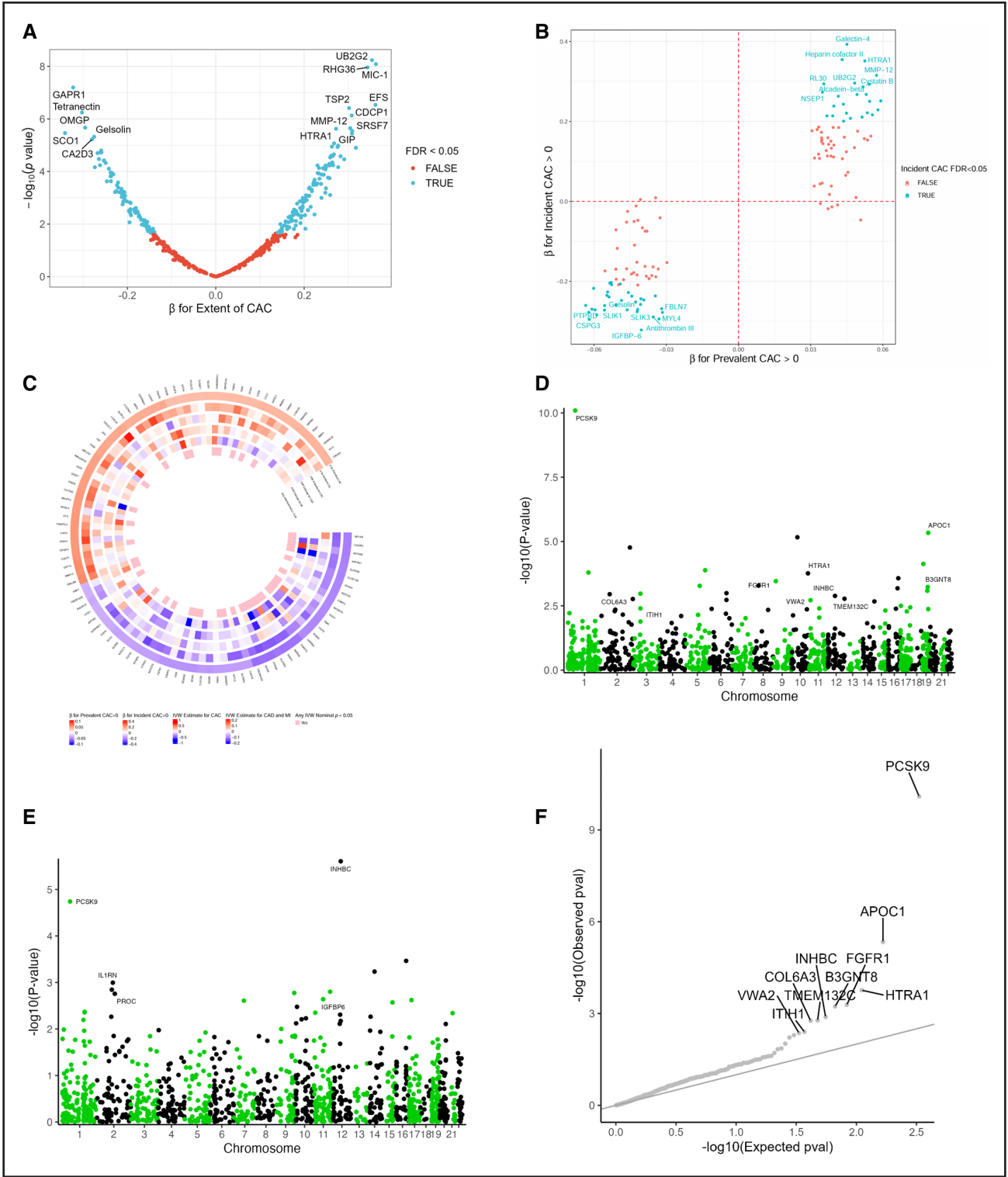


Figure 1. Circulating proteogenomic architecture of human coronary artery calcification. **A**, Volcano plot displaying the relation of aptamers with the prevalent extent of coronary artery calcium (CAC) [ln(CAC+1)] in the CARDIA study (Coronary Artery Risk Development in Young Adults) validation set (passed on from the derivation set, as described in Materials and Methods; see Tables 1 and 2 for derivation and validation split). Proteins significant at Benjamini-Hochberg false discovery rate (FDR) <0.05 are denoted in blue. **B**, Scatterplot depicting the relationship between 136 aptamers (associated with both presence and extent of CAC at baseline) and incident CAC>0 at follow-up. This analysis was conducted in 793 individuals free of CAC at baseline with proteomic and follow-up computed tomography data (see Materials and Methods). Aptamers associated with incident CAC are colored in blue. **C**, Circular heatmap depicting 100 proteins associated with both the presence and extent of CAC and with available protein quantitative trait loci as genetic instruments for Mendelian randomization analysis of CAC, coronary artery disease (CAD), and myocardial infarction (MI) (Continued)

plaque,⁷⁵ with successful PCSK9 interruption linked to lower vascular event rates.^{76,77} *APOC1* (lipoprotein metabolism) has been implicated in human GWAS or functional studies of atherosclerosis.^{78,79} *INHBC* (implicated in adipocyte metabolism) appears to link cardiometabolic health to coronary risk, with higher levels related to higher coronary risk.⁸⁰

Coronary Artery-Specific Transcriptome-Wide Association Studies (TWAS)

Circulating proteome-wide association does not resolve target tissue-specific mechanisms of disease pathogenesis, and circulating levels of proteins related to CAC may reflect systemic factors, not necessarily those operative within the vessel wall. TWAS in a mechanistically relevant tissue may address this limitation, estimating genetically determined transcript expression within a trait-related tissue (in this case, human coronary artery) to use in conjunction with cognate phenotype GWAS data (in this case, CAC). We constructed genetic models for gene expression in 268 human coronary arteries (see Materials and Methods), mapping these models to the largest extant GWAS of CAC²¹ to identify gene-level associations with CAC. Coronary artery expression of genes associated with CAC in TWAS (Figure 2A; full results in Table S6) included known (*PHACTR1*,⁸¹ *MRAS*,⁸¹ *MORF4L1-ADAMTS7*⁸²) and not previously widely implicated targets (*GIGYF1*, *DMPK*, *RPL9*). *PHACTR1* and *MRAS* demonstrated a strong association with CAC, consistent with pathogenic roles in endothelial dysfunction, CAC, or inflammatory cell function.^{81,83,84} Interestingly, *MORF4L1* and *ADAMTS7* are adjacent at 15q25.1 with the lead SNP in an intronic region of *MORF4L1*,⁸² leading to difficulty in resolving the actual causal disease gene. Although only *ADAMTS7* has wide support in models of atherosclerosis,⁸² our TWAS results (based on genetically determined RNA expression in the coronary artery), along with MR to account for linkage disequilibrium contamination, implicated both genes in CAC pathogenesis. Other genes identified in coronary TWAS of CAC displayed phenotypic association with anatomically or functionally adjacent processes, although not directly with atherosclerosis itself (*GIGYF1*: type 2

diabetes⁸⁵; *DMPK*: cardiac conduction⁸⁶; *RPL9*: ribosomal metabolism⁸⁷).

Over-representation analysis of top TWAS-implicated genes (Reactome; $P < 0.001$, 36 genes, Table S6) through WebGestalt⁸⁸ yielded nominally significant pathways ($P < 0.05$; Figure S2). Several displayed an enrichment ratio of >80 (calculated as the number of input genes in the pathway divided by the number expected based on the background protein coding genes), including HDL (high-density lipoprotein) clearance (notable given HDL's protective role against calcification⁸⁹), pre-NOTCH processing in the endoplasmic reticulum (dysfunction can shift vascular cells towards the calcification process⁶¹), and methionine salvage pathway (linked to coronary calcification via epigenetics⁹⁰).

Overlap Between TWAS and Proteomic Associations and Expression in Human Coronary Arteries

While proteins in circulation related to CAC may not reflect events in the coronary wall, we finally explored whether any overlap between the coronary artery-specific gene targets from TWAS and our CAC proteome associations would implicate genes of interest that may have roles in both coronary tissue and circulation. We identified 21 genes with support from circulating proteomic association with CAC in CARDIA as well as in human coronary artery TWAS of CAC (at a $P < 0.05$ threshold; Figure 2B; Table S7). Importantly, several of these genes have previously been implicated in vascular, metabolic or inflammatory mechanisms of atherosclerosis, including *PNPO* (candidate gene for hypertension⁹¹), *SPARCL1* (maintenance of vessel wall structure, lower in activated endothelial cells⁹²), *HES1* (responsible for cardiac arterial development⁹³), *S100A9* (vascular inflammation⁹⁴) and *ASGR1* (inhibition decreases hypercholesterolemia⁹⁵), and other known mediators of vascular dysfunction or plaque homeostasis (*NOTCH3*,⁹⁶ *TNFSF12*,⁹⁷ *S100A12*⁹⁸). In addition, while not previously directly implicated in CVD, *GPC6*, *RPP25*, and *HS6ST3* are mechanistically adjacent: *GPC6* blocks canonical Wnt signaling,⁹⁹ a pathway central to vascular calcification,^{100,101} and *RPP25* (a part of ribonuclease P) has

Figure 1 Continued. from genome-wide association studies. The heatmap illustrates the relationships between beta coefficients for proteomic associations with prevalent CAC >0 and incident CAC >0 , as well as inverse variance-weighted (IVW) effect sizes for CAC, CAD, and MI, demonstrating overall concordance. The innermost track highlights genes with IVW nominal $P < 0.05$ with any outcome (CAC, CAD, or MI). **D**, Manhattan plot for proteome-wide genetic association study (PWAS) of coronary atherosclerosis in UK Biobank (UK Biobank code: I9_CORATHER). **E**, PWAS for myocardial infarction in UK Biobank (UK Biobank code: I9_MI). Results in Table S6. For **D** and **E**, proteins that are labeled were associated with the presence and extent of CAC in CARDIA (derivation $P < 0.05$) and in the PWAS (for labeling purposes, labeled targets are $P < 5 \times 10^{-3}$). **F**, Quantile-quantile plot for the PWAS results for coronary atherosclerosis shown in **D**. This plot demonstrates enrichment (genomic inflation factor=1.460) for PWAS associations of genetically determined protein abundance with coronary atherosclerosis in the UK Biobank (UK Biobank code: I9_CORATHER) from circulating proteins associated with prevalent CAC presence and extent in CARDIA (derivation $P < 0.05$). EFS indicates embryonal Fyn-associated substrate; GIP, gastric inhibitory polypeptide; MMP, matrix metalloproteinase; PCSK9, proprotein convertase subtilisin/kexin type 9; SCO1, protein SCO1 homolog, mitochondrial; SRSF7, serine/arginine-rich splicing factor 7; and TSP-2, thrombospondin-2.

Table 3. Existing Molecular Evidence of Proteins With the Top 20 Absolute Effect Size for Extent of CAC in the Validation Sample That Are Associated With Both Presence and Extent of CAC

Gene/protein	CAC effect	Molecular evidence
Inflammation and fibrosis		
GDF-15	↑	A member of the TGF-β cytokine family; serves as a robust and independent biomarker for predicting disease progression and mortality in individuals with coronary artery disease ³⁷
EFS	↑	CRK-associated substrate adaptor protein; modulates T-cell function via negative selection; implicated in immune function and autoimmunity ³⁸
GSN	↓	Related to inflammation, lipid metabolism, and angiogenesis ³⁹ ; Expression is reduced in arteriosclerosis compared with normal tissue
CDCP1	↑	Regulates cardiac fibroblast proliferation, mitosis, and is implicated in myocardial fibrosis ⁴⁰
UBE2G2	↑	Involved in protein degradation; differentially expressed in early coronary atherosclerosis ⁴¹ and knockdown suppressed atherosclerotic plaque formation in mice ⁴²
GLIPR2	↓	Negative regulator of autophagy; upregulated in fibrosis ⁴³
Calcification		
ARHGAP36	↑	Involved in regulating bone development and metabolism; overexpression linked to heterotopic ossification ⁴⁴
RNAS6 (RNASE6)	↑	Expression increased in the blood and plaque of patients with cardiovascular disease; blocking RNAS6 may inhibit oxidized LDL-induced aortic smooth muscle cell migration and growth ⁴⁵
Extracellular remodeling		
CLEC3B (tetranection)	↓	Regulator of the fibrinolysis and proteolytic system; decreased circulating levels in patients with coronary artery disease ⁴⁶
MMP-12	↑	Involved in extracellular remodeling; overexpressed in atherosclerotic plaques and increased plasma levels associated with atherosclerotic burden and prevalent cardiovascular disease ^{47,48}
THBS2/TSP-2	↑	Modulates cell-matrix interactions; uncertain role in atherosclerosis development, TSP-2 mRNA is not expressed in the endothelial cells that line atheroma ⁴⁹
B4GALT6	↓	Related family members genetically implicated in cardiovascular risk ⁵⁰ ; important in the biosynthesis of lactosylceramide
Metabolism		
GIP	↑	Regulates vasodilation, vascular leukocyte adhesion, inflammation, and lipid metabolism; exerts both antiatherosclerotic and proatherosclerotic effects depending on the environment ⁵¹
Other		
SCO1	↓	Mitochondrial protein that regulates cellular copper importation; involved in oxidative stress response ⁵²
OMG	↓	Cell adhesion molecule required for myelination in the central nervous system ⁵³
CACNA2D3	↓	A component of the voltage-gated calcium channel complex
TMPO	↑	Plays a role in maintaining nuclear architecture and facilitating postmitotic nuclear reassembly
CSTB	↑	Inhibitor of cysteine proteases
TRAPPC3	↑	Plays a role in transporting proteins within the Golgi and endoplasmic reticulum ⁵⁴
SRSF7	↑	Regulate precursor RNA splice site recognition and spliceosome assembly; implicated in malignant cell proliferation ⁵⁵

ARHGAP36 indicates Rho GTPase-activating protein 36; B4GALT6, beta-1,4-galactosyltransferase 6; CAC, coronary artery calcium; CACNA2D3, voltage-dependent calcium channel subunit alpha-2/delta-3; CDCP1, CUB domain-containing protein 1; CLEC3B, tetranectin; CSTB, cystatin B; EFS, embryonal Fyn-associated substrate; GDF-15, growth/differentiation factor 15; GIP, gastric inhibitory polypeptide; GLIPR2, Golgi-associated plant pathogenesis-related protein 1; GSN, gelsolin; LDL, low-density lipoprotein; MMP, matrix metalloproteinase; OMG, oligodendrocyte-myelin glycoprotein; RNAS6, ribonuclease K6; SCO1, protein SCO1 homolog, mitochondrial; SRSF7, serine/arginine-rich splicing factor 7; TGF, transforming growth factor; THBS2/TSP-2, thrombospondin-2; TMPO, lamina-associated polypeptide 2, isoforms beta/gamma; TRAPPC3, trafficking protein particle complex subunit 3; and UBE2G2, ubiquitin-conjugating enzyme E2 G2.

been implicated in genetic studies of cardiometabolic disease¹⁰² and in scleroderma (a disease marked by multiorgan calcification and fibrosis). Genetic variants in *HS6ST3*—a gene whose product has been implicated in protein–extracellular matrix interactions—have been linked to CAC in some settings,¹⁰³ although implications in human atherosclerosis have not been investigated. Finally, these 21 genes were expressed across broad cell types in human coronary arteries,²⁹ with evidence of differential expression in key cell types between atherosclerotic and nonatherosclerotic coronary tissues (eg, *S100A9* higher in macrophages in atherosclerotic tissue,

with several lines of evidence across our entire study; Figure 2C through 2D). Results for pseudobulk differential expression are in [Figure S3](#).

Examination of our quantile-quantile plots suggested that 8 genes (*NOTCH3*, *SPINK2*, *S100A12*, *RPP25*, *OAF*, *HS6ST3*, *TNFSF12*, and *GPC6*) exhibited the greatest divergence from expectation from our overall 21 genes (Figure 2B). These 8 genes were therefore prioritized for final MR approaches and phenome-wide association studies to assess whether these targets would have genetic support for a role in coronary calcification (Figure S4). We performed inference via MR

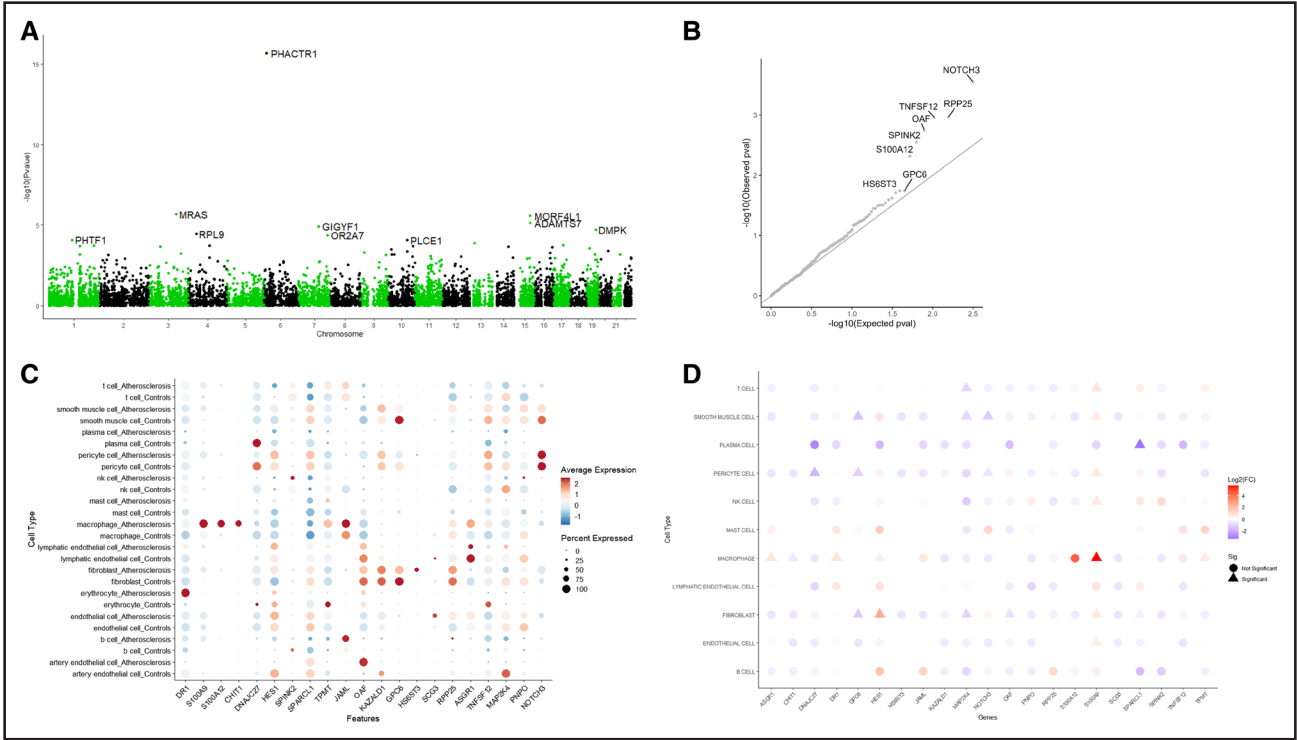


Figure 2. Human coronary transcriptome-wide association study (TWAS) of coronary artery calcium (CAC) and its overlap with proteomic targets from the CARDIA study (Coronary Artery Risk Development in Young Adults). **A**, TWAS of CAC using genetic models of transcript expression in human coronary artery samples from GTEx (genotype-expression tissue database; see Materials and Methods). **B**, Quantile-quantile plot showing enrichment (genomic inflation factor = 1.059) for coronary artery-specific TWAS associations for CAC, where we have tested proteins associated with prevalent presence and extent of CAC in CARDIA (derivation $P < 0.05$). The top 8 transcripts are labeled. **C** and **D**, Results of single-cell RNA sequencing in human coronary arteries. **C**, Distribution of expression by condition (atherosclerotic or nonatherosclerotic coronary artery tissue) and cell type. **D**, Differential expression across cell types for atherosclerotic vs nonatherosclerotic coronary artery tissue.

with gene expression as exposure and CAC as outcome. For causal inference, we used 5 MR approaches, each relying on strong assumptions on the underlying ground truth: MR-Egger,¹⁰⁴ median-based,¹⁰⁵ maximum likelihood,¹⁰⁶ inverse variance weighted,¹⁰⁷ and MR-JT²⁸ (Figure S4A). For 5 of these genes (*SPINK2*, *RPP25*, *OAF*, *HS6ST3*, *GPC6*), the direction of effect observed in MR analyses was consistent with the direction of the circulating proteomic association with CAC. In phenome-wide association studies across the heritable phenotype in UK Biobank (heritability $h^2 P < 0.05$) using genetically determined expression in the coronary artery, many of the 8 prioritized genes implicated vascular, inflammatory, and obesity-related phenotypes (Figure S4B).

DISCUSSION

Here, we unite population-wide circulating proteomics, noninvasive coronary imaging, and circulating and coronary artery tissue-specific genomics to present a framework for prioritizing targets for CVD susceptibility and downstream mechanistic work. Our studies within a large population sample (CARDIA) indexed a broad human proteome to prevalent and incident CAC—an

independent marker of subclinical disease and incident clinical coronary artery disease and death before age 60 years¹¹—implicating pathogenic mechanisms across oxidative stress and lipid metabolism, extracellular matrix remodeling, immune cell function, coagulation, and inflammation.^{7,108,109} Given our rationale that coronary artery-specific contextualization may focus discovery for CVD, we next performed the largest human coronary artery TWAS of CAC, mapping bulk transcriptomics in 268 human coronary arteries to the largest CAC GWAS, demonstrating enrichment for genes implicated by proteomic associations across similar pathways of metabolic-inflammatory activation expressed in key cell types relevant to coronary disease progression. These 2 results—PWAS and TWAS of human CAC—have not been previously reported to our knowledge and are provided here as a discovery resource. Finally, while we recognize that genes implicated in CAC from coronary artery tissue may be distinct from circulating proteomic targets, we explored targets with evidence across both modalities using MR and single-cell transcriptomics, identifying several targets with multilevel support or prior biological curation relevant to vascular disease. Collectively, these results furnish a framework to decipher disease-relevant

targets through integration of human genetic approaches with multiomics.

In addition to well-known mechanisms of inflammation and fibrosis (Table 3), proteomics identified directionally consistent targets with strong evidence in model systems (less in humans), including *TREM2* (attenuates macrophage uptake of oxidized lipids, limiting progression of atherosclerosis in mice⁶³), *Notch1* (loss potentiates *RUNX2*-mediated calcification⁶¹), and *ARHGAP36* (overexpression associated with connective tissue to bone transformation⁴⁴), among others (Table 3). Importantly, we observed a strong concordance between effect estimates for prevalent and 10-year incident CAC in CARDIA, with the strongest agreement for both known (eg, MMP-12, GDF-15) and emerging mediators of vascular risk. Genetics (MR) suggested directional consistency in genomic-proteomic effects for many targets, many of which have not been extensively implicated in human coronary disease by GWAS studies but have relevant metabolic-vascular mechanisms (eg, *HS6ST3*,¹¹⁰ *PTPRD*,¹¹¹ *C1QTNF1*,⁷¹ *DNJB9*¹¹²). PWAS approaches (genomic predictions of the proteome) were also supportive, identifying one of the strongest therapeutic targets in coronary disease (PCSK9), now with support in primary prevention for vascular events (in line with our genetic findings).⁷⁷

A second new finding of our approach is deploying a TWAS based on human coronary artery transcription (from GTEx) in the largest GWAS of CAC. This TWAS approach has recently been used to identify targets essential to cardiometabolic therapeutics,¹¹³ though its application to coronary arteries has—to our knowledge—not been reported. The current report represents the largest TWAS of CAC in human coronary arteries to date. Human coronary TWAS associations with CAC included known (*PHACTR1*,⁸¹ *MRAS*,⁸¹ *MORF4L1-ADAMTS7*⁸²) and not widely implicated targets (*GIGYF1*, *DMPK*, *RPL9*). In addition, pathway analysis of TWAS hits identified pathways putatively involved in calcification, including HDL clearance and NOTCH. Given the high concordance of therapeutic targets based on genetic supportive evidence,¹¹⁴ we think this data may provide an important starting point for further downstream mechanistic studies to inform potential therapeutic targets.

In addition to an unbiased TWAS approach, we explored targets implicated both by proteomic associations with CAC as well as in the CAC TWAS. This approach has some precedent in recent studies linking the human proteome to tissue-specific transcriptional states,^{115–120} although we certainly recognize that events in the coronary wall may not necessarily be reflected in the circulating proteome (and vice versa). The overlap between the coronary artery TWAS of CAC and the circulating proteomics of CAC (from CARDIA) highlighted 21 genes, some with known implications in vascular, inflammatory, or metabolic mechanisms in model systems (*NOTCH3*,⁹⁶

TNFSF12,⁹⁷ among others) and some not previously studied. Certainly, lack of concordance between circulating protein abundance and coronary tissue expression may limit discovery, and proteins in circulation may reflect systemic processes that influence CAC development. Nevertheless, convergent support from previous studies, proteomics, and coronary tissue TWAS highlights these as biologically plausible candidates for further mechanistic and therapeutic evaluation. This fundamental observation highlights the importance of studying both the peripheral circulation (where targets like PCSK9 may be found in proteogenomic approaches) as well as the coronary tissue itself as parallel, orthogonal approaches to coronary disease discovery.

Our study has several important limitations. Importantly, our analyses proceeded in parallel (proteomic and proteogenomic studies in goal 1 and genetic-transcriptomic studies in goal 2; Figure 1) in recognition that CAC susceptibility is driven by systemic, peripheral factors as well as coronary-specific factors. Our discovery of new proteins not quantified on this platform (Somalogic) is limited by design and can be addressed as proteomic platforms advance in coverage. Further, coronary arteries in our genomics approaches may not necessarily reflect the range of coronary phenotypes relevant to disease discovery, and we were not able to ascertain the exact histological classification of the coronary arteries used in TWAS from GTEx. Although differences in directionality of effect between tissue MR approaches and protein association are possible, our goal was to identify potential targets for downstream interrogation. Although our circulating proteomic studies utilized standard type 1 error control methods (FDR), we utilized more lenient thresholds in our genomics analyses, given that these were meant to be multiple lines of supportive evidence beyond the circulating proteome. Also, additional efforts across proteogenomic approaches that involve multiple ancestries are critical to understanding the basis for coronary disease susceptibility across populations, given then European ancestry predominant cohorts used here. Finally, mechanistic studies are needed to provide biological validation and final prioritization of targets for therapeutic interruption. Nevertheless, multilevel discovery in tissues and populations provides a focus for calcification mechanisms and biology for wider studies and therapeutic approaches.

In conclusion, we present an approach to address coronary artery calcification susceptibility across the circulating proteome, coronary artery-specific transcription, and TWAS, implicating genes both known and not previously reported in CAC pathogenesis. Targets identified by our approach enjoyed multilevel support across proteogenomic approaches, several with previously reported biological implications in CVD or CVD-adjacent mechanisms. This approach represents a multidimensional approach to human coronary atherosclerosis, incorporating both circulating and tissue-derived marks of disease

liability to identify multilevel determinants of coronary disease for future mechanistic studies.

ARTICLE INFORMATION

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Disclosures

A.S. Perry is a coinventor on proteomic signatures of fitness, liver, and respiratory disease. S. Das is a founder and owns equity for Thryv Therapeutics and Switch Therapeutics and is a consultant for Thryv Therapeutics. He has had research grants from Bristol Myers Squibb and Abbot Laboratories. E.R. Gamazon performs consulting for Thryv Therapeutics. R.V. Shah is supported by grants from the National Institutes of Health. R.V. Shah has equity ownership in and is a consultant for Thryv Therapeutics. R.V. Shah is a coinventor on pending patents or disclosures on molecular biomarkers of fitness, lung disease, cardiovascular diseases and phenotypes, and metabolic health, use of RNAs as therapeutics and diagnostic biomarkers in disease, and methods in metabolomics. E.R. Gamazon is a coinventor on pending patents or disclosures on molecular biomarkers of cardiovascular and metabolic diseases and phenotypes, use of RNAs as therapeutics and diagnostic biomarkers in disease, and methods in metabolomics. The other authors report no conflicts.

Supplemental Material

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